EMNLP2026 BUDAPEST

KOREA UNIVERSITY · NAVER Cloud · KAIST AI Kim Jaechul Graduate School · FURIOSA

Text Capability Loss in Vision-Language Adaptation: An Attention-Sink Diagnosis

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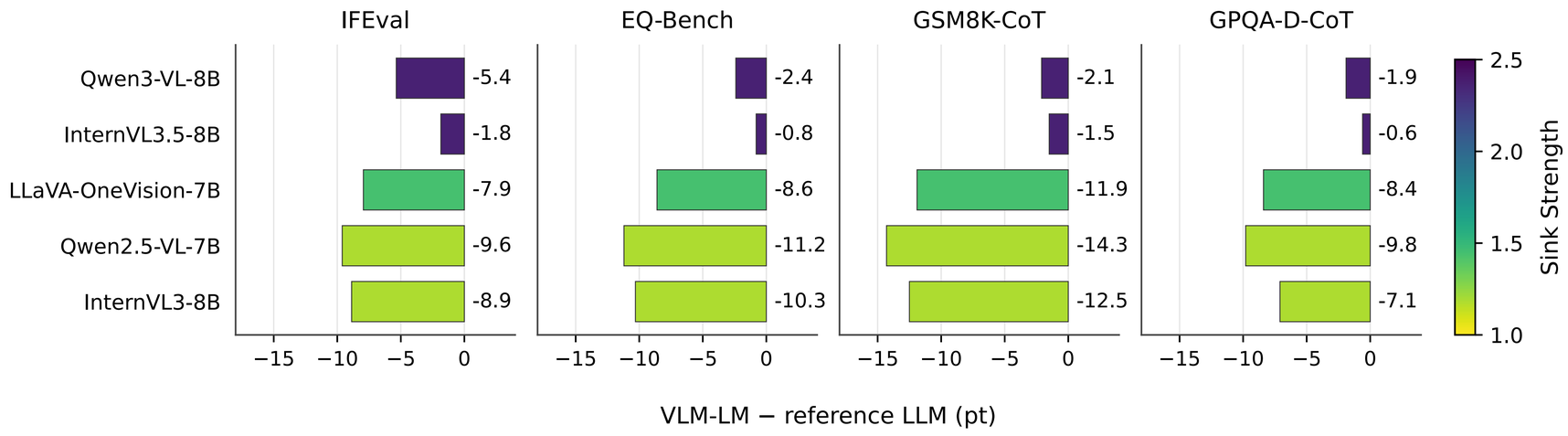
[1]Korea University · [2]NAVER Cloud AI · [3]KAIST AI · [4]FuriosaAI

Paper · Code

Turning an LLM into a VLM degrades instruction following.

Sink Strength, one scalar on the base LLM, predicts the loss before any VL training.

1 VL adaptation erodes format-sensitive text capability



−14.3 pt
largest drop on format-sensitive tasks

≤ 2.3 pt
drop on knowledge tasks (MMLU, BoolQ)

0.0 vs −9.6 pt
IFEval gap: text-only training vs VL training

INSTRUCTION · IFEval

Answer in ALL CAPS. Use no commas.

paraphrased from the IFEval prompt

BASE LLM · Qwen2.5-7B-Instruct — PASS

"… THE PRACTICE OF USING NATURAL FEATURES TO CREATE A MORE SUSTAINABLE ENVIRONMENT"

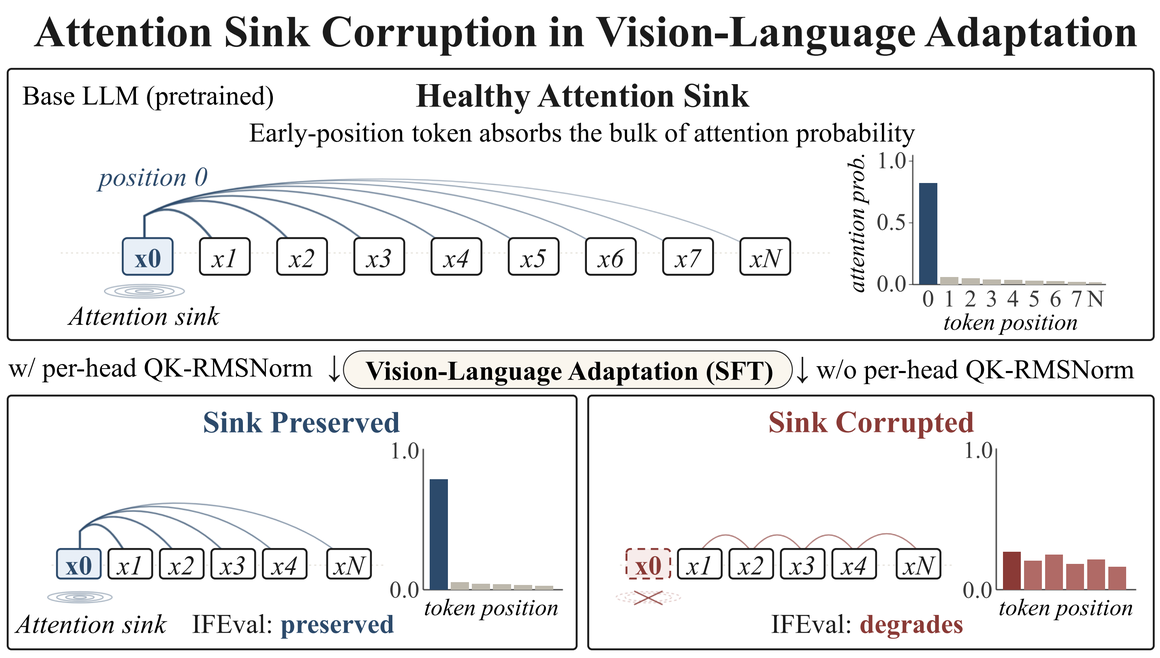
no commas

AFTER VL TRAINING · Qwen2.5-VL-7B — FAIL

"… IT PROMOTES BIODIVERSITY, SUSTAINABLE HABITATS, AND ECOLOGICAL SERVICES …"

uses commas

2 Why: VL fine-tuning corrupts the attention sink



Sink. An early token that absorbs most attention.

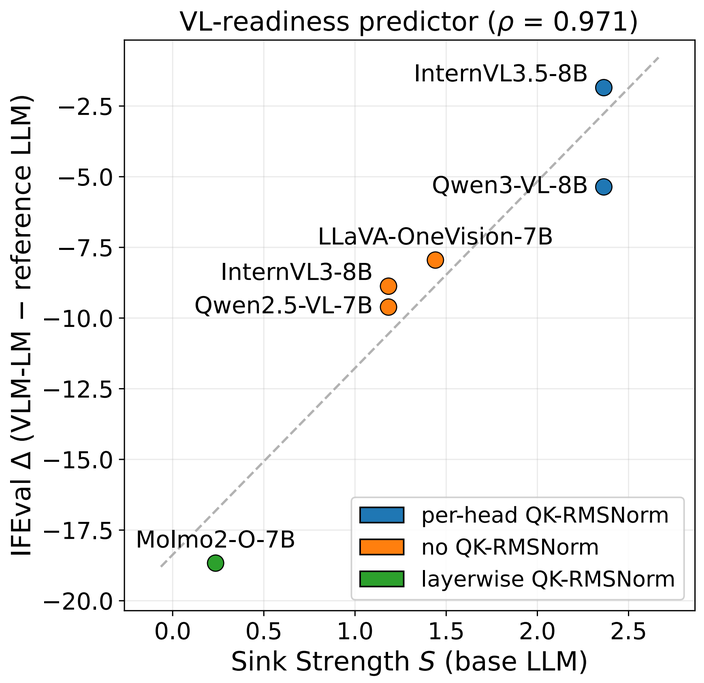
Damage. VL updates shift attention scores, amplified at the sink.

Protection. Per-head QK-RMSNorm removes it (Theorem 1).

3 Sink Strength S: screen before VL training

HOW S IS MEASURED

1. Run the base LLM on 15 text prompts
2. For each attention head: a = attention on the sink
3. S = median over heads of log(a / (1 − a))



ρ = 0.971
S vs IFEval gap, six pairs (ρ = 0.88 on 17 pairs)

2.54 pt
leave-one-pair-out error (mean baseline 4.40 pt)

ρ ≥ 0.88
on each of the four format-sensitive tasks

~8 s
per 7B LLM on a single GPU, inference only

4 Granularity matters: per-head vs layerwise

BASE-LLM DESIGN	SINK STRENGTH S	IFEVAL GAP
Per-head QK-RMSNorm Qwen3-VL, InternVL3.5	2.36	−1.8 to −5.4
No QK-RMSNorm Qwen2.5-VL, InternVL3, LLaVA-OV	1.18–1.44	−7.9 to −9.6
Layerwise QK-RMSNorm Molmo2-O (control)	0.24	−18.7

A "has QK-norm?" check passes Molmo2-O. S flags it as the highest risk.

5 Finding: simple fixes do not undo the damage

Adding QK-RMSNorm later does not protect

11 vs 59.9

IFEval: injected model (first checkpoint) vs base LLM

Merging back toward the LLM does not recover it

0 of 7

merging settings recover IFEval

Takeaway: screen the base LLM with Sink Strength before VL training.